ECON 5345 Homework 3 Report

Harlly Zhou

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Question 1

The FRED said that the TWEXBPA is no longer updated since some time around 2015, and it started to use RTWEXBGS "Real Broad Dollar Index" to substitute. So I used the RTWEXBGS instead.

Table 1: Unit root / stationarity tests on log levels (test statistic and critical values).

Series	Test (spec.)	Stat	CV(1%)	CV(5%)	CV(10%)
Panel A: log(RTWEXBGS), monthly (N=242)					
RTWEXBGS	PP (Z-τ, const.)	-0.9725	-3.4589	-2.8736	-2.5731
RTWEXBGS	PP (Z-τ, trend)	-2.8287	-3.9993	-3.4297	-3.1381
RTWEXBGS	ADF (τ_2, drift)	-1.2374	-3.4600	-2.8800	-2.5700
RTWEXBGS	ADF (τ_3, trend)	-3.2276	-3.9900	-3.4300	-3.1300
RTWEXBGS	KPSS (μ)	3.7950	0.7390	0.4630	0.3470
RTWEXBGS	KPSS (τ)	0.5219	0.2160	0.1460	0.1190
Panel B: log(GDPC1), quarterly (N=316)					
GDPC1	PP (Z-τ, const.)	-2.2491	-3.4528	-2.8709	-2.5717
GDPC1	PP (Z-τ, trend)	-1.3899	-3.9908	-3.4256	-3.1357
GDPC1	ADF (τ_2, drift)	-2.3118	-3.4400	-2.8700	-2.5700
GDPC1	ADF (τ_3, trend)	-1.5289	-3.9800	-3.4200	-3.1300
GDPC1	KPSS (μ)	5.3256	0.7390	0.4630	0.3470
GDPC1	KPSS (τ)	1.0925	0.2160	0.1460	0.1190

Question 2

a. Note that we can write

$$\operatorname{Var}(e) = [\Gamma_{ij}]_{i,j=1}^{n} \in \mathbb{R}^{Tn \times Tn},$$

where $\Gamma_{ij} = \operatorname{Cov}(e_i, e_j) \in \mathbb{R}^{T \times T}$ is the (i,j)-th block of $\operatorname{Var}(e)$. Since v_t is white noise, we have

$$\operatorname{Cov}(e_{it}, e_{j\tau}) = \begin{cases} \Omega_{ij} \in \mathbb{R}, & t = \tau \\ 0, & \text{otherwise} \end{cases}$$

Therefore, we have $\Gamma_{ij} = \operatorname{diag}(\Omega_{ij}) = \Omega_{ij} I_T$. Hence,

$$\operatorname{Var}(e) = \Omega \otimes I_T.$$

b. Since Ω is a covariance matrix, it is positive definite. Hence, Ω is non-singular. By the properties of Kronecker product, we have

$$\begin{aligned}
\hat{\beta}_{\text{GLS}} &= \left((I_n \otimes X')(\Omega^{-1} \otimes I_T)(I_n \otimes X)\right)^{-1} \left((I_n \otimes X')(\Omega^{-1} \otimes I_T)y\right) \\
&= (\Omega^{-1} \otimes X'X)^{-1}(\Omega^{-1} \otimes X')y \\
&= \left(\Omega \otimes (X'X)^{-1}\right)(\Omega^{-1} \otimes X')y \qquad \text{since } X'X \text{ is positive-definite} \\
&= \left(I_n \otimes (X'X)^{-1}X'\right) y \\
&= \left(I_n \otimes (X'X)^{-1}\right)(I_n \otimes X')y \\
&= \left(I_n \otimes (X'X)^{-1}\right)((I_n \otimes X)'y) \\
&= ((I_n \otimes X)'(I_n \otimes X))^{-1}((I_n \otimes X)'y) \\
&= \hat{\beta}_{\text{OLS}}.
\end{aligned}$$

Question 3

a. 1